OLIVIA A. M. MARTIN

Curriculum Vitae

Email — Website — LinkedIn — Google Scholar

RESEARCH INTERESTS & EXPERTISE

- Water-mineral interfaces, H-bonding structure and dynamics, computational spectroscopy
- Membrane-mineral & membrane-water interfaces, prebiotic chemistry, origins of life, thermodynamics of membrane formation
- Computational & Theoretical Methods: Molecular dynamics (MD) simulations, path integral MD, umbrella sampling and indirect umbrella sampling (INDUS), statistical mechanics, nuclear quantum effects, machine learning (ML)
- Software & Hardware Skills: Linux, Bash, Shell Scripting, GROMACS, LAMMPS, i-PI, PLUMED, high-performance computing (HPC), Python (NumPy, SciPy, PyTorch)

EDUCATION

Ph.D. in Chemistry, Physical Chemistry — Expected May 2027
Advisor: Dr. Richard C. Remsing — Rutgers University, Piscataway, NJ

B.S. in Chemistry, *cum laude* — May 2022
Advisor: Dr. Nathan C. Eddingsaas — Rochester Institute of Technology, Henrietta, NY
Minor: Environmental Sciences

RESEARCH EXPERIENCE

Graduate Student Researcher — June 2022 - Present
Advisor: *Dr. Richard C. Remsing*
Department of Chemistry and Chemical Biology, Rutgers University, Piscataway, NJ

Emerson Summer Undergraduate Research Fellowship — May - August 2021
Advisor: *Dr. Nathan C. Eddingsaas*
College of Science, Rochester Institute of Technology, Henrietta, NY

Undergraduate Student Researcher — August 2018 - May 2022
Advisor: *Dr. Nathan C. Eddingsaas*
Department of Chemistry and Materials Science, Rochester Institute of Technology, Henrietta, NY

PUBLICATIONS

Peer-Reviewed Publications

1. "A bellwether for microplastic in wetland catchments in the Great Lakes region" Dominique S. Gilbert, Brett A. Hayhurst, Sarah Grubisich, Nick Schneider, **Olivia Martin**, Christopher DeNyse, Kristina M. Chomiak, Anna Christina Tyler, Nathan C. Eddingsaas. Journal of Great Lakes Research, **2024**, 50, 102411 DOI: 10.1016/j.jglr.2024.102411

Manuscripts In Preparation

3. "Electric Fields on Short Chain Lipids at the Water-Alumina Interface" **Olivia A. M. Martin**, Richard C. Remsing. Target Journal: J. Phys. Chem. B, (In Preparation)

2. "Nuclear Quantum Effects at the Alumina/Water Interface" Somaiyeh Dadashi, Ziyad Thekkayil, **Olivia A. M. Martin**, Aashish Tuladhar, Richard C. Remsing, Eric Borguet. Target Journal: Nature Chemistry, (In Preparation)

1. "Enhanced Sampling Molecular Dynamics Simulation Methods to Predict the Critical Micelle Concentration" **Olivia A. M. Martin**, Richard C. Remsing. Target Journal: Physical Review Letters, (In Preparation)

CURRENT AND PAST FUNDING

External Grants

Funded

Proposal Title: Astrobiology Graduate Conference (ABGradCon) 2026
Funding Agency: National Aeronautics and Space Administration
Role Status: Collaborator
Percentage of Attributed Effort: 20
Total direct costs and annual budget: $119,640
Coverage Period: June - December 2026

HONORS AND AWARDS

2022 Undergraduate Senior Chemistry Achievement Award, *American Institute of Chemists*
2021 Nathaniel Rochester Society Scholarship
2021 Good Citizenship Award, *Rochester Institute of Technology*
2020 Nathaniel Rochester Society Scholarship
2020 Good Citizenship Award, *Rochester Institute of Technology*
2020 Alumni Endowed Academic Scholarship, *College of Science, Rochester Institute of Technology*
2020 Outstanding Second Year Student Award, *Rochester Institute of Technology*

SELECTED TALKS AND PRESENTATIONS

Oral Presentations

7. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α-Alumina Interface", Water and Aqueous Solutions Gordon Research Seminar 2026, Holderness, NH

6. "Molecular and Thermodynamic Insights into Prebiotic Lipid Assembly", Astrobiology Science Conference 2026, Madison, WI

5. "Lipid Assembly Formation on Minerals: A Path to Protocells", Astrobiology Graduate Conference 2025, Boulder, CO

4. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Graduate Conference 2024, Ithaca, NY

3. "Interfacial Structure and Vibrational Relaxation of Aqueous Solutions near Alumina", American Chemical Society Mid-Atlantic Regional Meeting (MARM) 2024, State College, PA

2. "Analytical Methods and Processes of Environmental Microplastic", RIT Undergraduate Research Symposium 2021, Rochester, NY

1. "The chemical and physical changes to plastics aged in surface and benthic freshwater aquatic systems", State of Lake Ontario Conference 2021, Rochester, NY

Poster Presentations

9. "Molecular and Thermodynamic Insights into Water and Amphiphiles at the α-Alumina Interface", Water and Aqueous Solutions Gordon Research Conference 2026, Holderness, NH

8. "Thermodynamic Characteristics of Water and Amphiphiles at the Alumina Interface", Rutgers Chemical Physics Symposium 2026, Piscataway, NJ

7. "Structure and Dynamics of Water and Amphiphiles at the α-Alumina Surface", Chemistry and Physics of Liquids Gordon Research Conference 2025, Holderness, NH

6. "Structure and Dynamics of Water and Amphiphiles at the α-Alumina Surface", Chemistry and Physics of Liquids Gordon Research Seminar 2025, Holderness, NH

5. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Chemistry in Solution and at Interfaces Workshop: Deep Modeling for Molecular Simulations Workshop 2024, Princeton, NJ

4. "Fatty Acid Assembly at Water-Mineral Interfaces", Water and Aqueous Solutions Gordon Research Conference 2024, Holderness, NH

3. "Fatty Acid Assembly at Water-Mineral Interfaces", Water and Aqueous Solutions Gordon Research Seminar 2024, Holderness, NH

2. "Thermodynamics of Primitive Lipid Assemblies at Mineral Surfaces", Astrobiology Science Conference 2024, Providence, RI

1. "Microplastic Pollution in the Benthos of Rural and Urban Tributaries of Southern Lake Ontario", Joint Aquatic Sciences Meeting 2022, Grand Rapids, MI

TEACHING AND ADVISING

Graduate Teaching Assistant 2024 - Present
Organic Chemistry I Lab Rutgers University, Piscataway, NJ

Graduate Mentor 2024 - 2025
Remsing Lab Rutgers University, Piscataway, NJ

Tutor 2022 - 2023
Organic Chemistry I and II Rutgers University, Piscataway, NJ

Undergraduate Mentor 2021 - 2022
College of Science Rochester Institute of Technology, Henrietta, NY

Undergraduate Teaching Assistant 2020 - 2022
General and Analytical Chemistry Lab Rochester Institute of Technology, Henrietta, NY

Learning Assistant 2020 - 2021
General Chemistry I and II Rochester Institute of Technology, Henrietta, NY

PROFESSIONAL SERVICE AND LEADERSHIP

GRS Co-chair 2027
Chemistry and Physics of Liquids Gordon Research Seminar 2027 Holderness, NH

Rutgers Chemical Physics Early-Career Committee 2026 - Present
Rutgers University, Piscataway, NJ

Conference Co-Organizer 2025 - Present
Astrobiology Graduate Conference 2026 University of Arizona, Tucson, AZ

Director of Fundraising and Communications 2025 - Present
Astrobiology Graduate Conference 2026 University of Arizona, Tucson, AZ

GRS Discussion Leader 2025
Chemistry and Physics of Liquids Gordon Research Seminar 2025 Holderness, NH

Lab Manager 2024 - Present
Remsing Lab Rutgers University, Piscataway, NJ

Director of Outreach 2023 - 2025
Chemistry & Chemical Biology Graduate Student Organization Rutgers University, Piscataway, NJ

Treasurer, Executive Board 2023 - 2025
Supporting Parents and Caregivers at Rutgers Organization Rutgers University, Piscataway, NJ

Volunteer 2019 - 2022
Women in Science Rochester Institute of Technology, Henrietta, NY

Young Chemists Committee 2020 - 2021
American Chemical Society Rochester Institute of Technology, Henrietta, NY

President 2019 - 2021
American Chemical Society, RIT Chapter Rochester Institute of Technology, Henrietta, NY

Student Ambassador 2018 - 2021
College of Science Student Advisory Board Rochester Institute of Technology, Henrietta, NY

Volunteer 2020
Rochester Museum and Science Center Rochester, NY

Event Coordinator 2019
American Chemical Society Rochester Institute of Technology, Henrietta, NY

Hospitality Chair 2019
64th Annual Undergraduate Research Symposium, American Chemical Society Rochester, NY

REFERENCES

Richard C. Remsing, PhD
Associate Professor
Department of Chemistry and Chemical Biology, Rutgers University
123 Bevier Rd, Piscataway, NJ 08854
Phone: (848) 445-3245 | Email: rick.remsing@rutgers.edu
Relationship: Primary Ph.D. Advisor

Lu Wang, PhD
Associate Professor
Department of Chemistry and Chemical Biology, Rutgers University
123 Bevier Rd, Piscataway, NJ 08854
Phone: (848) 445-4555 | Email: lwang@chem.rutgers.edu
Relationship: Thesis Committee Member

Paul G. Falkowski, PhD
Board of Governors Professor; Director, Rutgers Energy Institute
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Relationship: Thesis Committee Member

Harender Singh Dhattarwal, PhD
Postdoctoral Research Associate
Department of Chemical and Biomolecular Engineering, Cornell University
113 Ho Plaza, Ithaca, NY 14853
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Relationship: Mentor; Postdoctoral Research Associate, Remsing Lab